Fire Finder
Uncertainty Presentation

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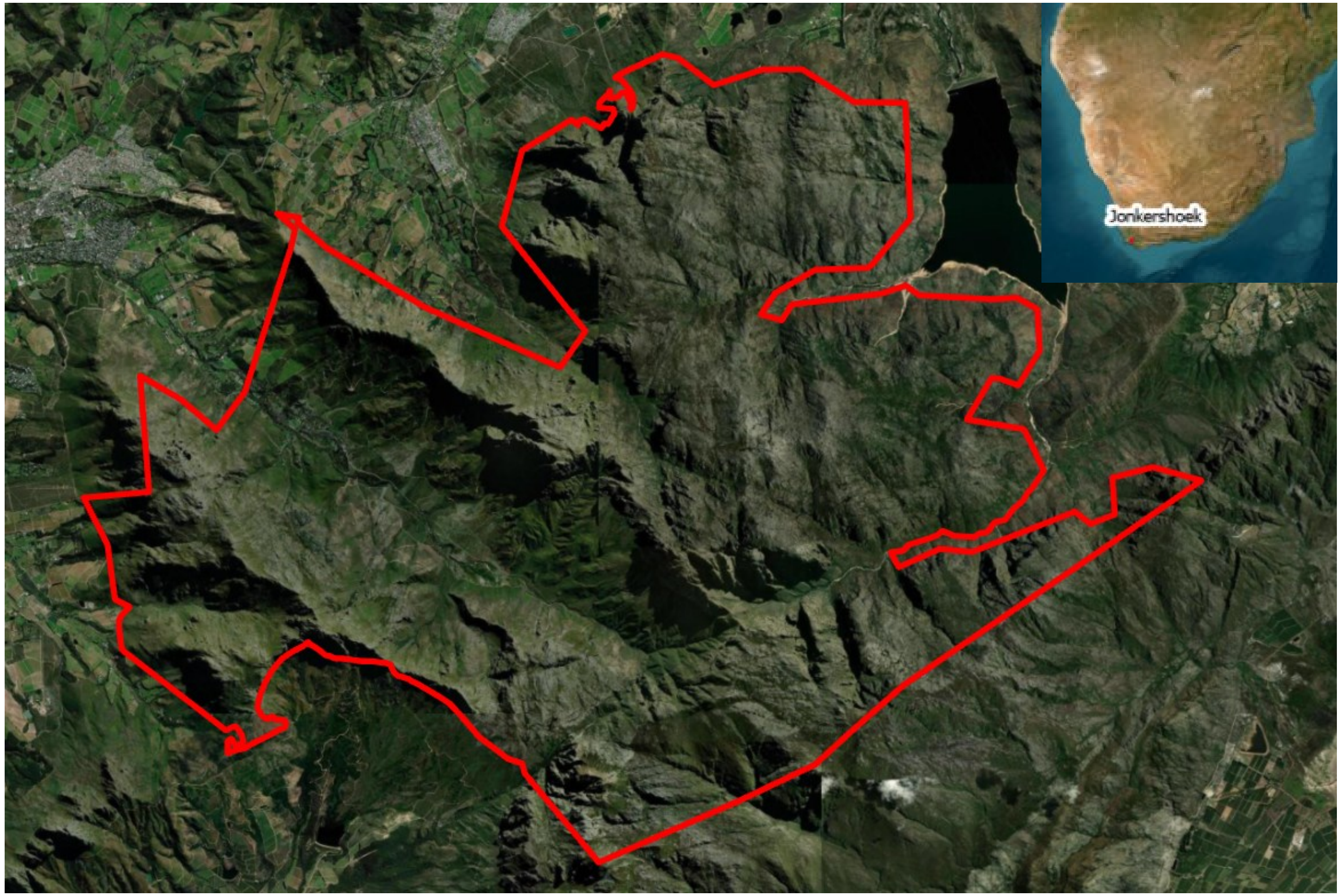
Study area and fire susceptibility context



- Jonkershoek Nature Reserve, 10 000 ha of mountainous fynbos.
- Mediterranean climate, steep terrain, fire-adapted fynbos = seasonal fire regime.
- High-stakes location: headwaters of the Eerste & Berg Rivers; directly adjacent to Stellenbosch CBD and farms.
- Recent fires exposed a gap in proactive risk management - CapeNature need a defensible susceptibility map to guide firebreaks, alien clearing, and response planning.

Variable selection and justification

Physiographic

Slope drives fire spread velocity; aspect controls solar exposure (north-facing slopes drier in the Southern Hemisphere); TPI captures ridge vs. valley position, with ridges being windier and more fire-prone *(Pourtaghi et al., 2015).*

Biotic Fuel

NDVI is a proven biomass and fuel-load proxy in fynbos specifically, and land cover differentiates flammability between fynbos, thicket, and invasive aliens *(Wilson et al., 2015).*

Human Ignition

The majority of wildfires in the Cape Floristic Region are anthropogenic, so proximity to roads and settlements is a direct proxy for ignition probability (Forsyth & Van Wilgen, 2008).

Historical

Fynbos has a fire return interval of 10–15 years, so time since last burn is a direct proxy for fuel accumulation and renewed susceptibility (Van Wilgen et al., 2010; Kraaij et al., 2013).

Criterion standardisation methods

Reclassification: All raw rasters were reclassified to a common ordinal scale of 1 (Very Low) to 5 (Very High Susceptibility).

Normalization: Continuous variables like Slope were binned using literature-defined thresholds consistent with South African Fynbos fire behavior literature. The variables without expert-defined thresholds were standardised using Natural Jenks Algorithm.

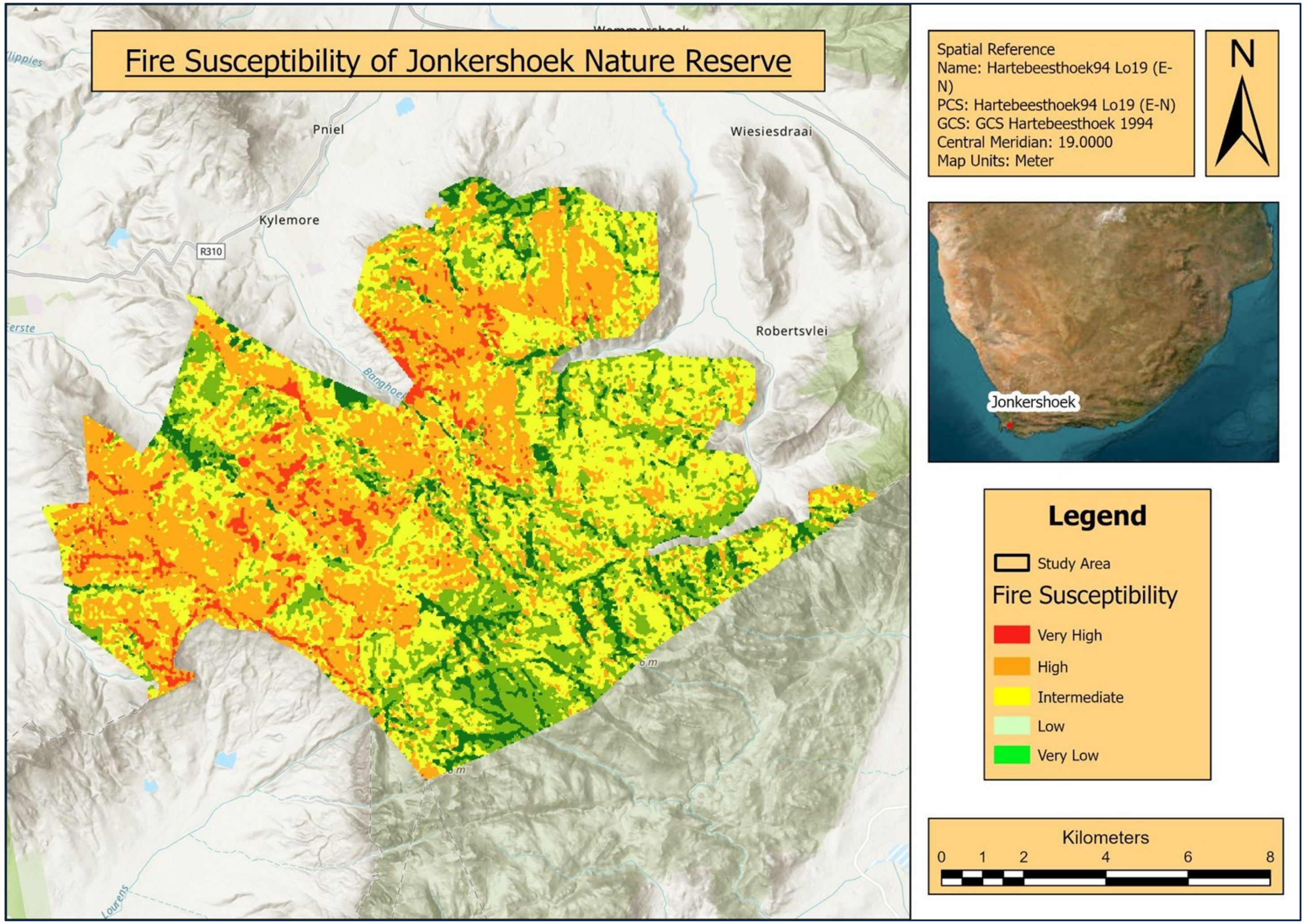
Spatial Uniformity: All layers were snapped to a 30m cell size resolution, using Bilinear Interpolation for topographic features to maintain surface continuity.

Transparency: This common scale allows for weighted additive combination, ensuring no single unit of measurement dominates the model.

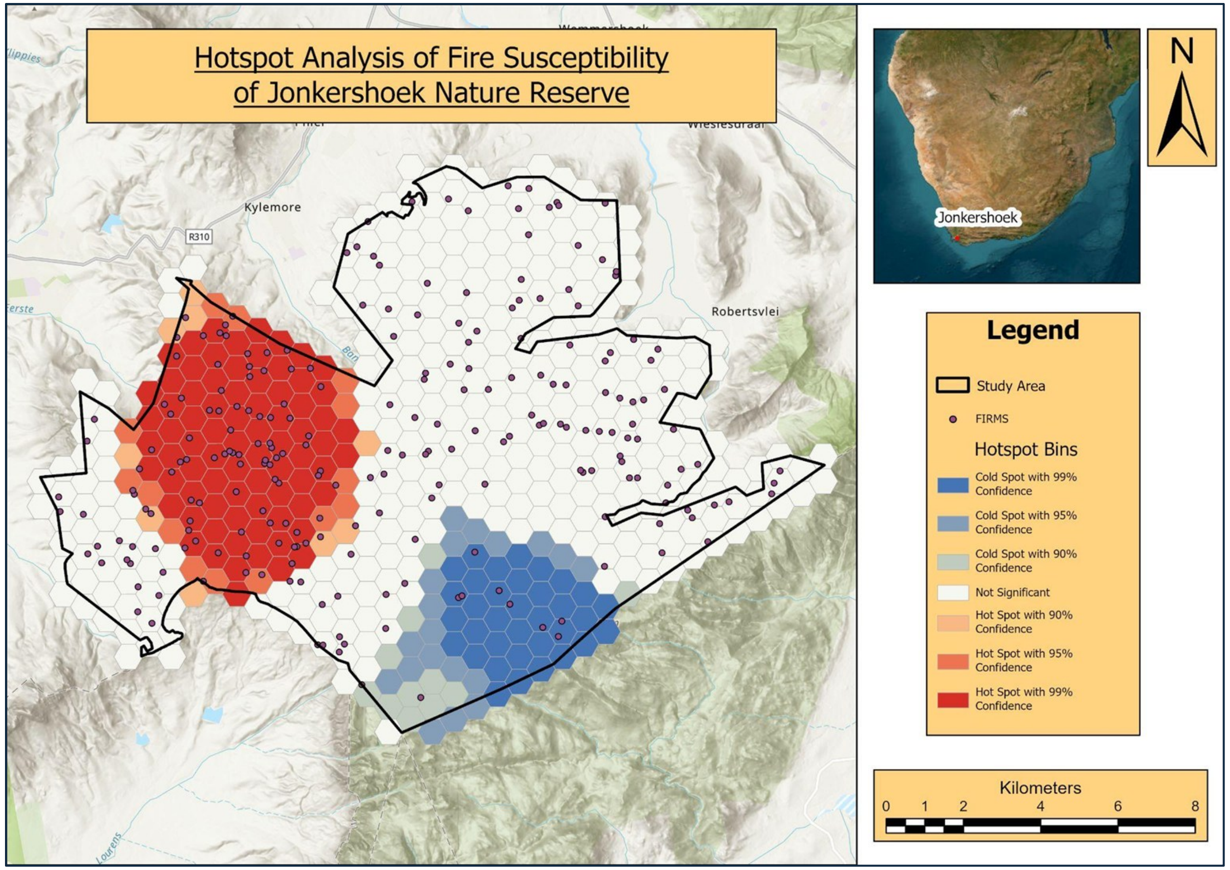
Weighting approach

Feature	Base Weight (%)	Physical Justification
NDVI	15.0	Biomass density and fuel load
TPI	15.0	Acts as a Wetness and Elevation Index
Land Cover	15.0	Structural fuel connectivity
Settlement	13.0	Accidental human ignition
Historical Fire	10.0	Long-term recurring areas
Slope	8.0	Convection and radiation
Roads	8.0	Access corridors
Aspect	6.0	Solar desiccation of fuel

The Final Fire Susceptibility



Validation approach - FIRMS hotspot analysis



Hotspot Analysis

The Method: Aggregated historical NASA FIRMS data into hexagonal bins to identify statistically significant spatial clustering of fire events.

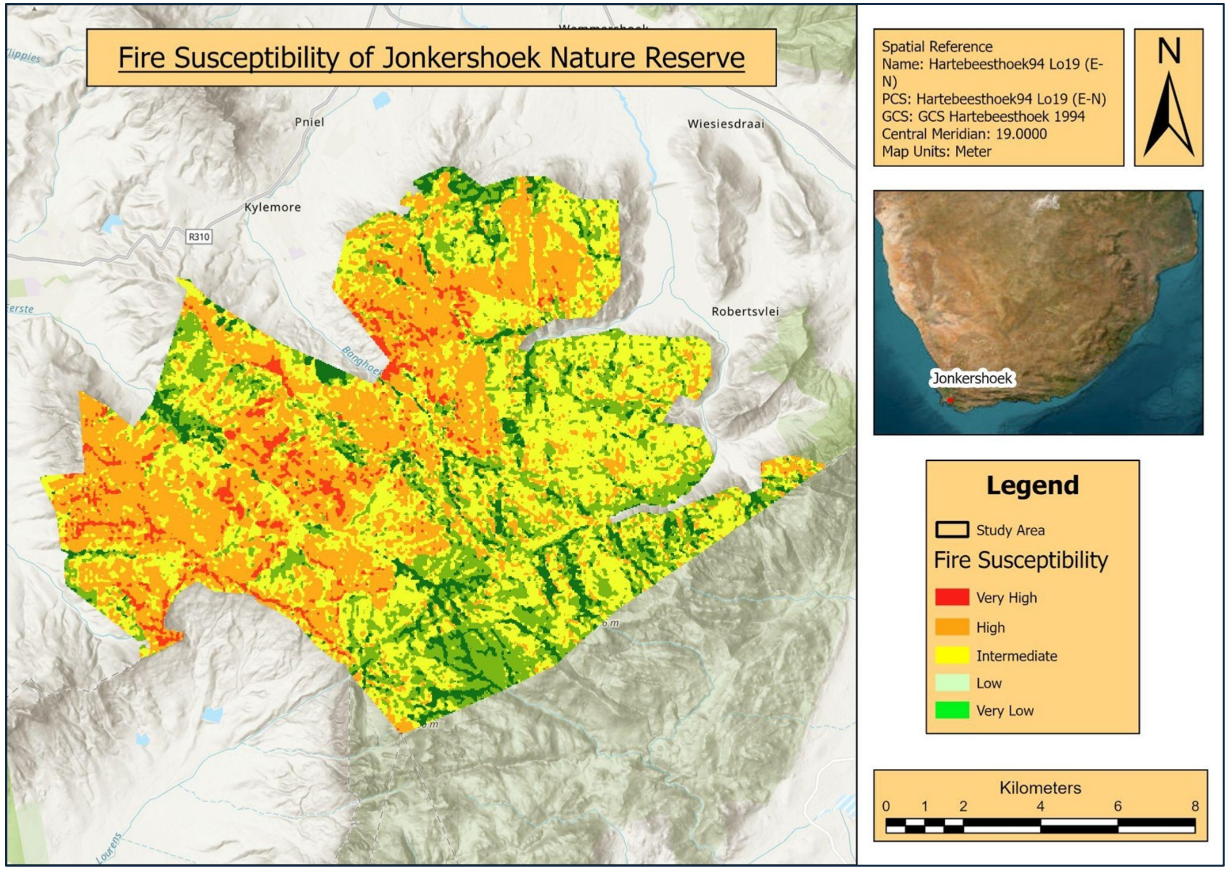
Why Hotspot Analysis? Hotspot analysis was used because it statistically identifies significant spatial clusters of fire events rather than relying on visual inspection alone (Getis & Ord, 1992). FIRMS is an appropriate independent validation dataset since it was not used as a model input.

The Result: Strong spatial correlation. The 99% confidence Hot Spot (NW sector) directly overlays the MCE's "Very High" susceptibility zones, while the 99% Cold Spot (S sector) aligns with the modeled "Low" risk zones.

Critical Limitations:

- *Scale Mismatch*: Comparing 375m resolution satellite thermal anomalies against a high-resolution (~30m) topographical MCE limits micro-scale validation.
- *MAUP Susceptibility:* The exact boundaries of the hotspot validation are influenced by the selected hexagonal bin size.
- *Phenomenological Difference:* FIRMS captures realized fire events (often wind-driven), whereas the MCE models static environmental potential.

Validation approach - FIRMS hotspot analysis



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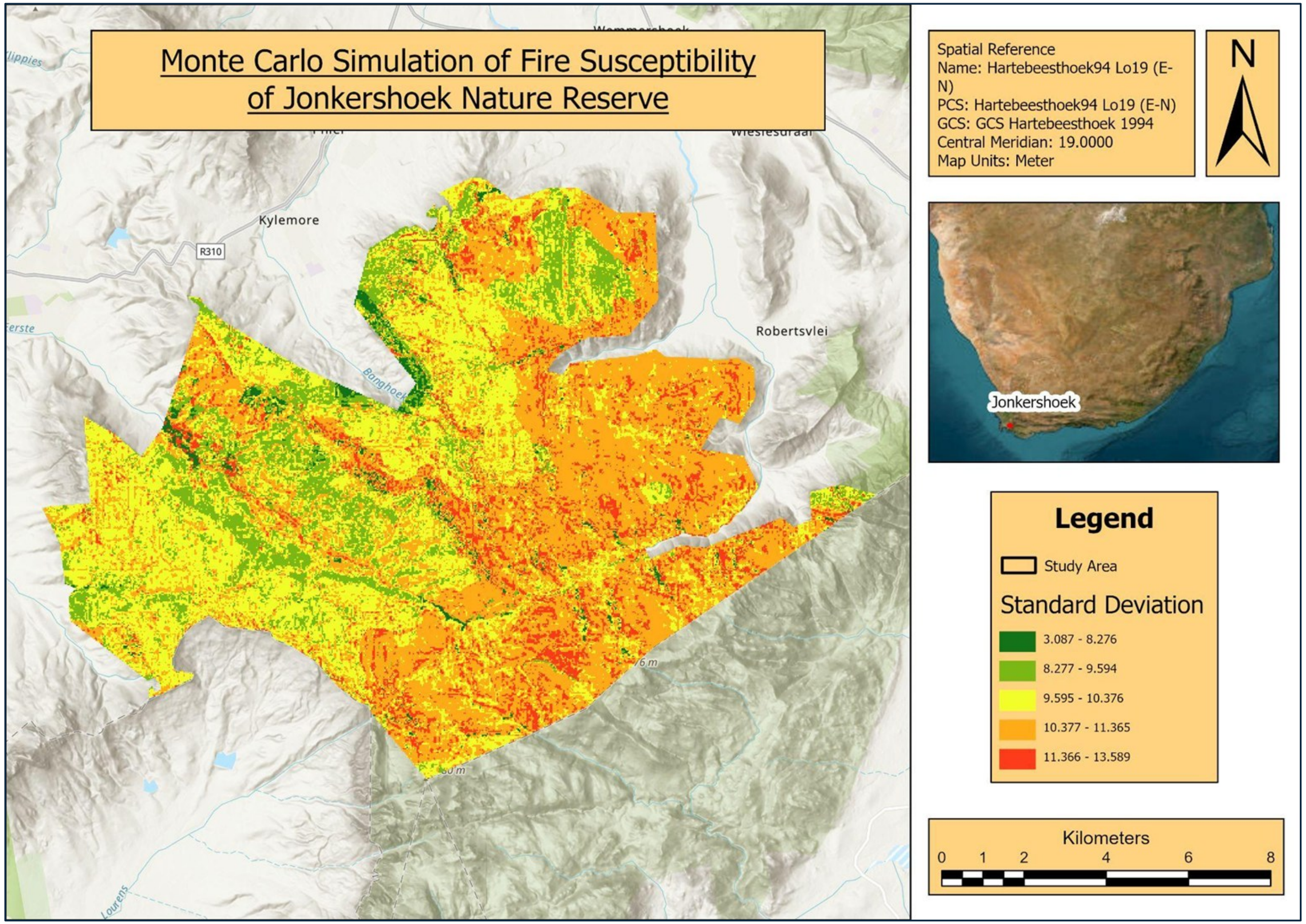
Sensitivity and uncertainty analysis - Monte Carlo Simulation

Monte Carlo Method

- **Why?** Quantifies how subjective weighting decisions propagate into MCE outputs (Feizizadeh et al., 2014). The ± 20% perturbation simulates expert disagreement, while re-normalisation preserves the original weighting constraint.
- **Execution:** 100 automated iterations using ArcPy Map Algebra.
- **Perturbation:** 8 base criteria weights were dynamically randomized (±20% variance) and re-normalized per run to maintain a constant sum of 90.
- **Output:** Standard Deviation calculated per pixel.
- **Result:** Green zones confirm highly robust risk predictions; Red/Orange zones indicate high sensitivity to subjective weighting bias.



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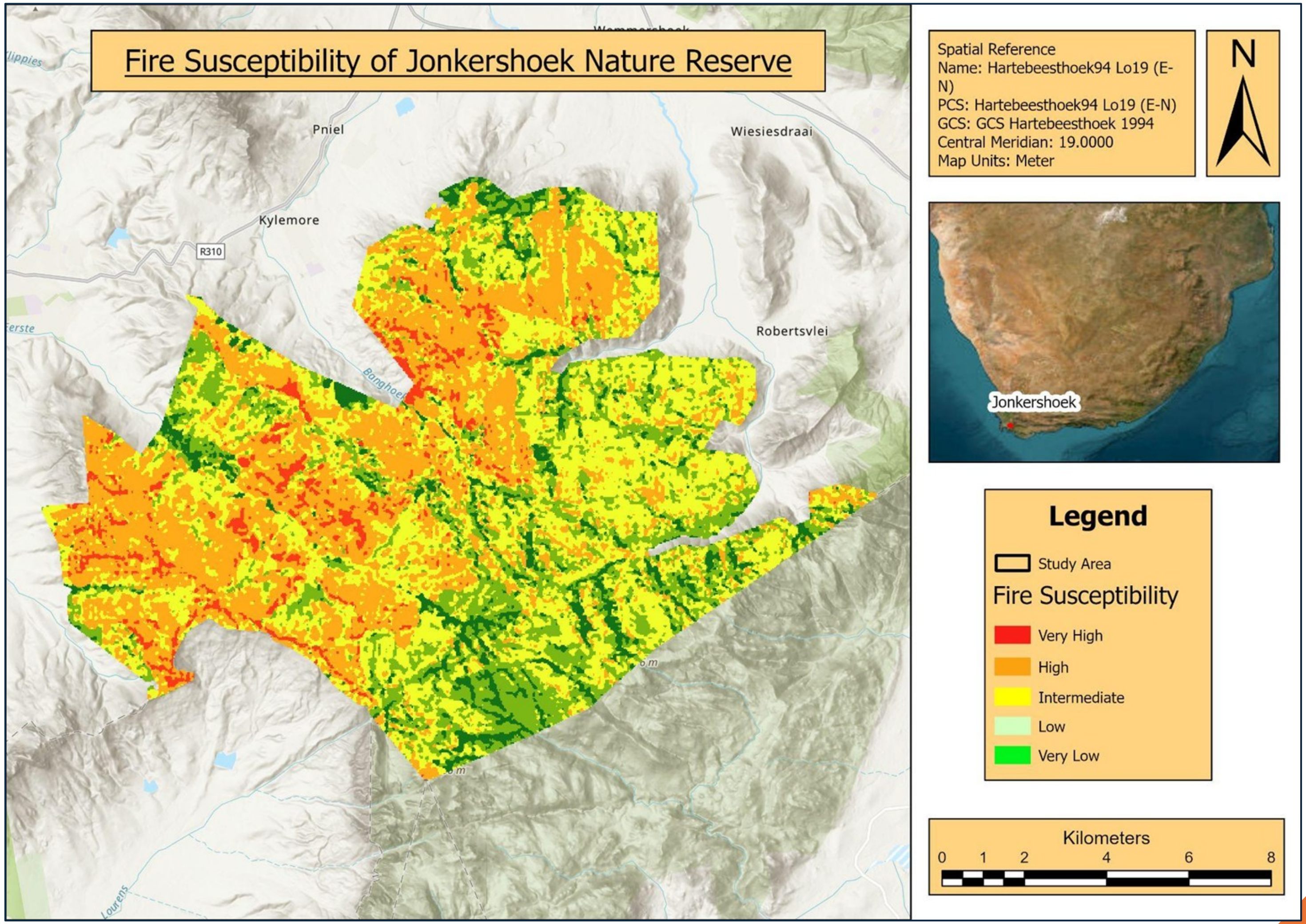
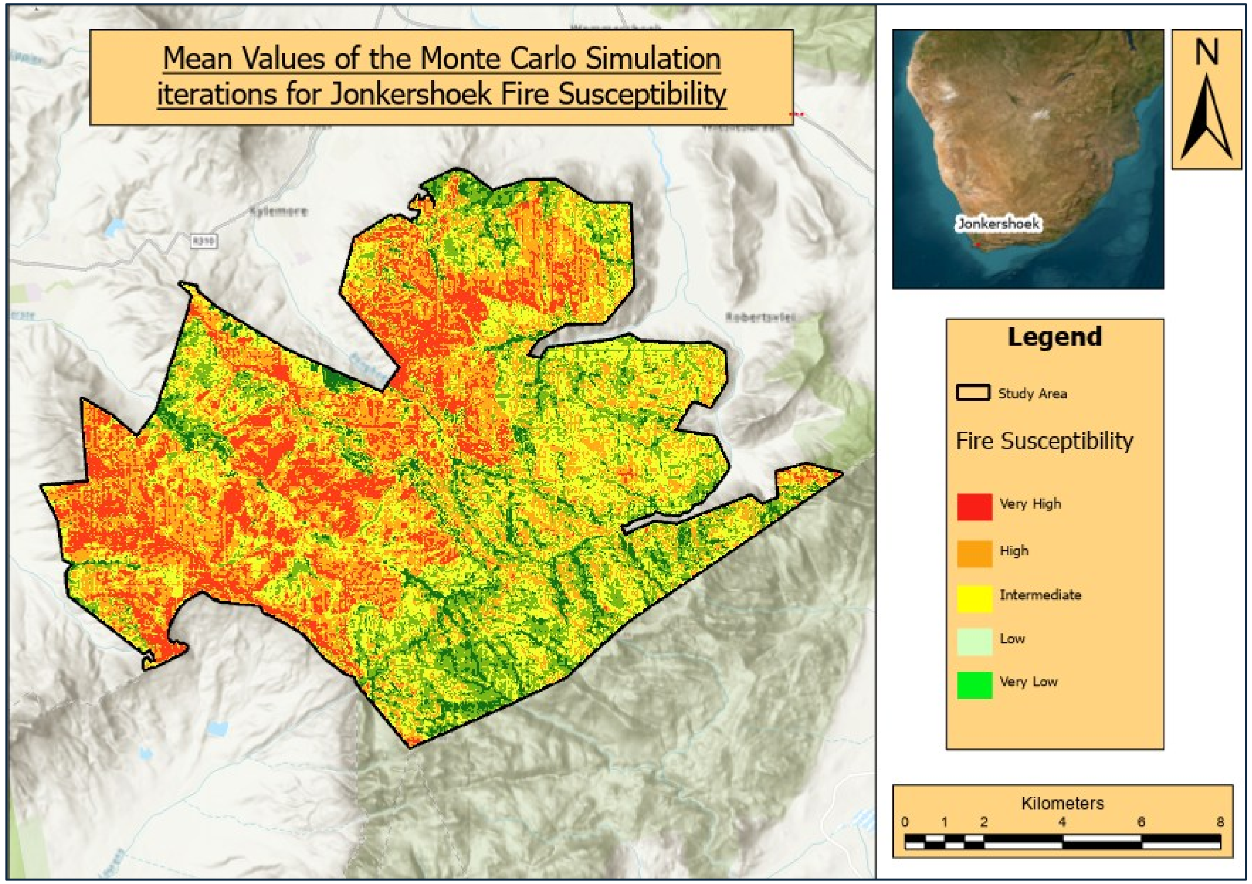
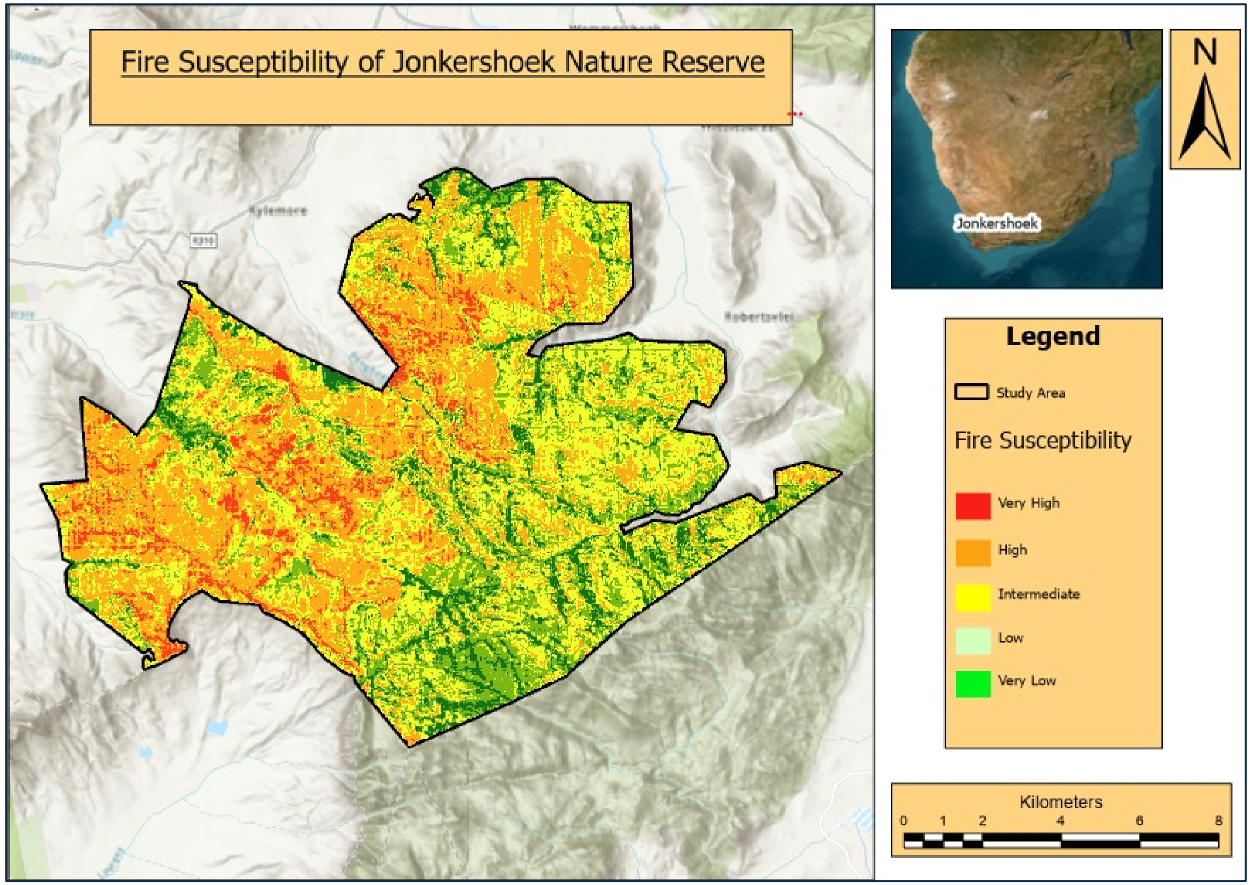
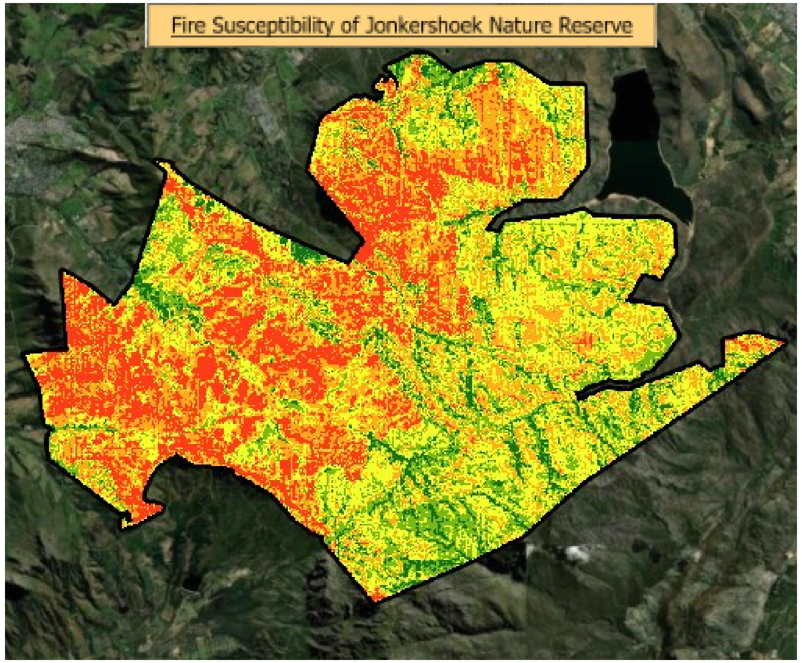


Sensitivity and uncertainty analysis
- Monte Carlo Simulation

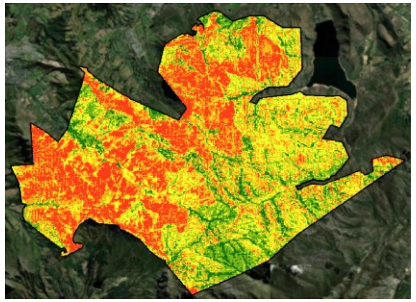




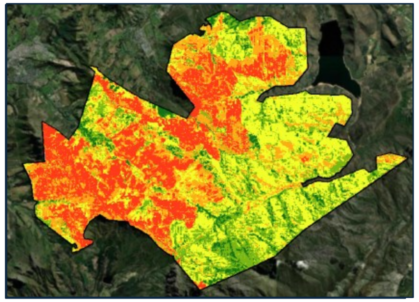
Sensitivity and uncertainty analysis
- Jack Knife Analysis



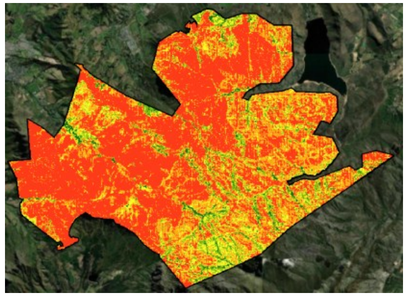
Without Aspect



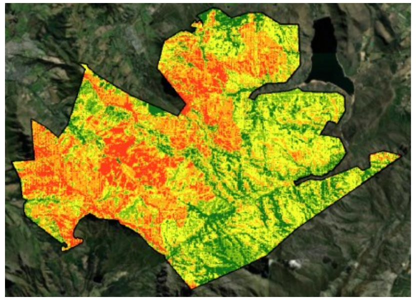
Without TPI



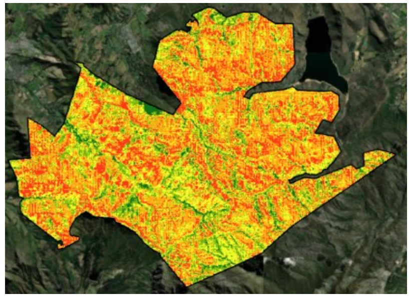
Without Slope



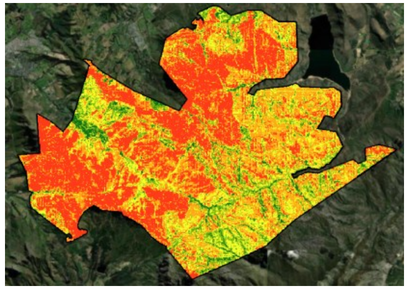
Without Settlement



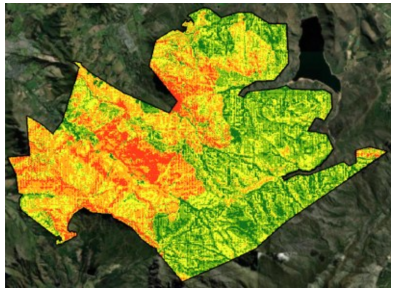
Without Historical Fires



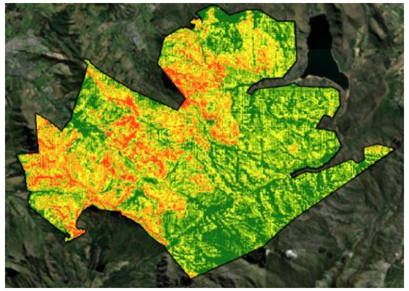
Without Roads



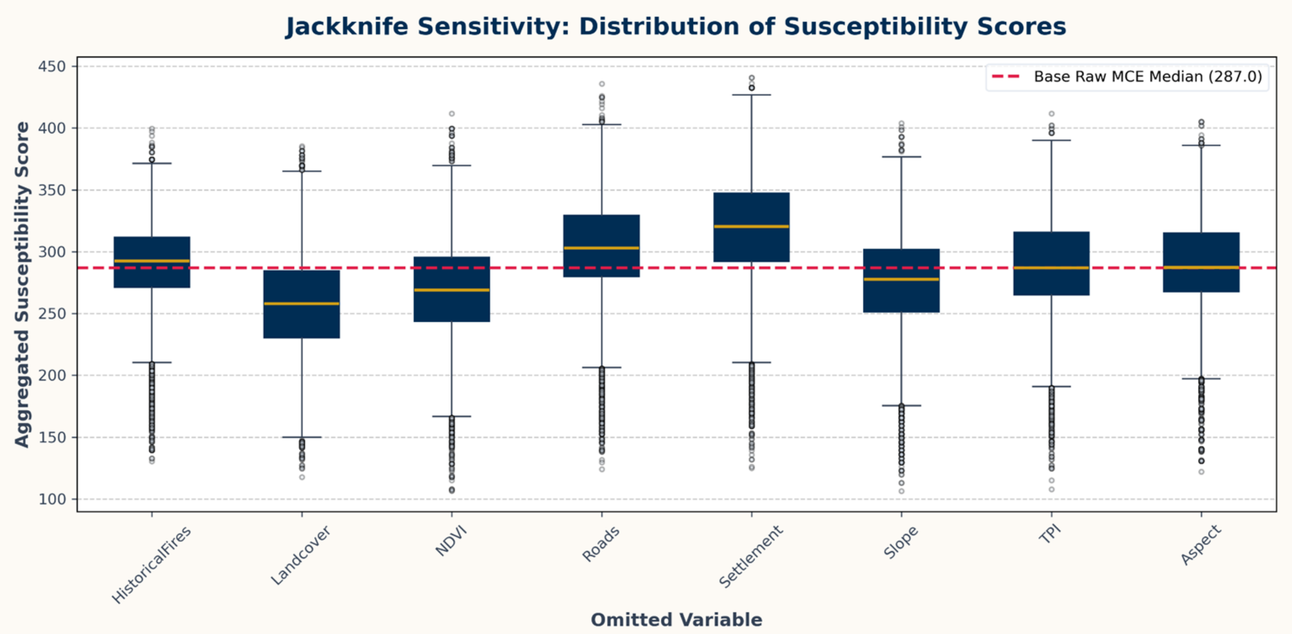
Without NDVI



Without Land cover

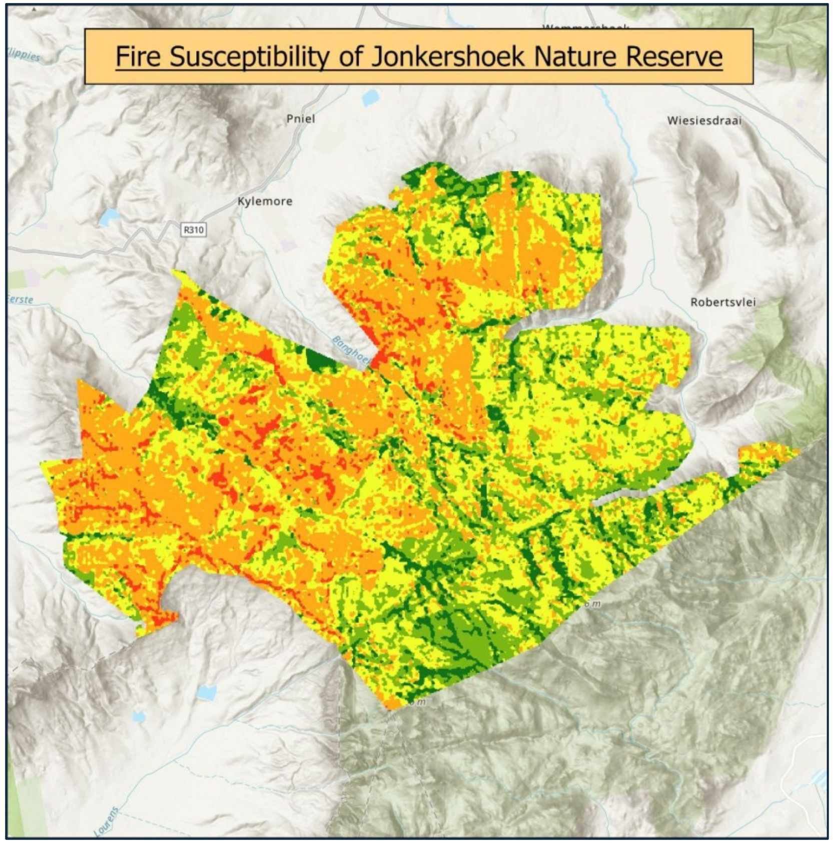


Stability or uncertainty quantification

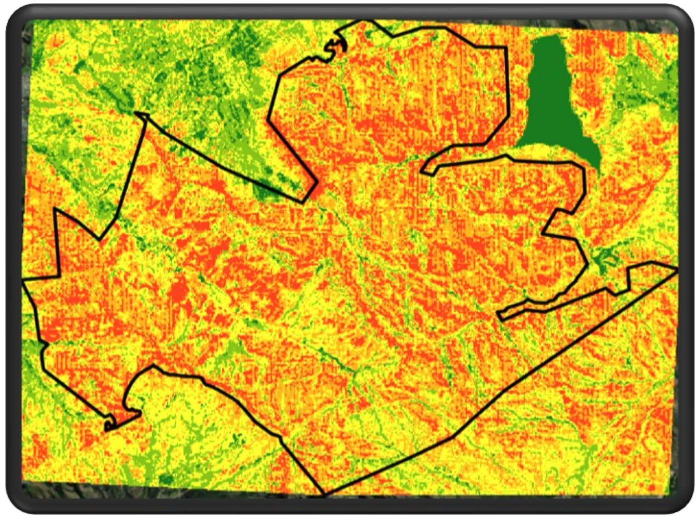


Impact	Variables	Direction	Interpretation
Critical	**Settlement** (+33), **Landcover** (−32)	Positive / Negative	The two pillars. Settlement suppresses risk; Landcover explains it. Omitting either breaks the model.
High	**Roads** (+18), **NDVI** (−17)	Positive / Negative	Secondary protective and risk factors. Roads add spatial structure; NDVI captures vegetation flammability.
Low	**Slope** (−7)	Negative	Weak independent effect; largely redundant.
Negligible	**Historical Fires** (+3), **TPI** (0), **Aspect** (0)	Positive / No Effect	Model compensates perfectly. Aspect is the least important variable.

Sensitivity and uncertainty analysis
-   Jack Knife Analysis



Without Historical Fires

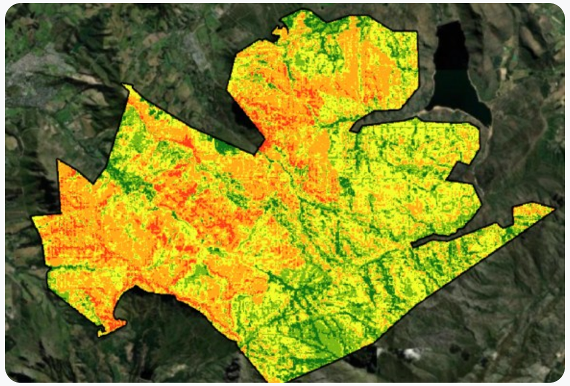


- **Why jackknife?** Jackknife analysis isolates each variable's contribution by removing it one at a time, identifying which inputs most strongly drive the model output *(Phillips et al., 2006)*.

- **Key finding:** Removing Historical Fires caused the largest shift, susceptibility was substantially over-predicted across the reserve, confirming historical fire frequency as an influential variable. This aligns with fynbos fire ecology, where time-since-burn is the dominant control on fuel load *(Van Wilgen et al., 2010)*.

Sensitivity and Uncertainty Analysis

Impact of Spatial Resolution

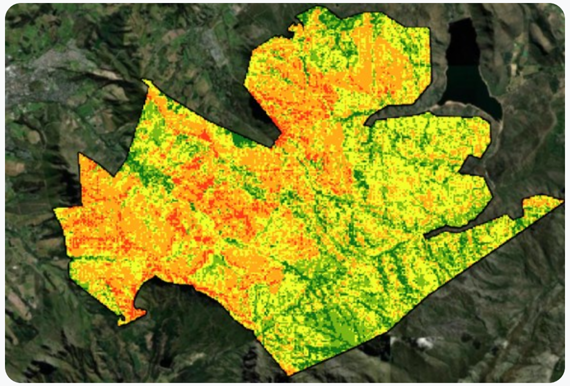


15m

Spatial Resolution

Highest level of detail capturing micro-scale topographical variations.

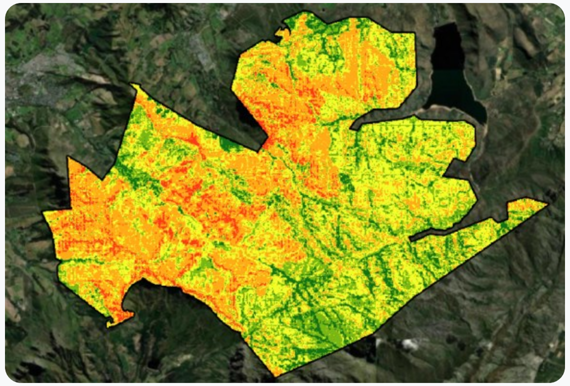
May introduce noise into the model



60m

Spatial Resolution

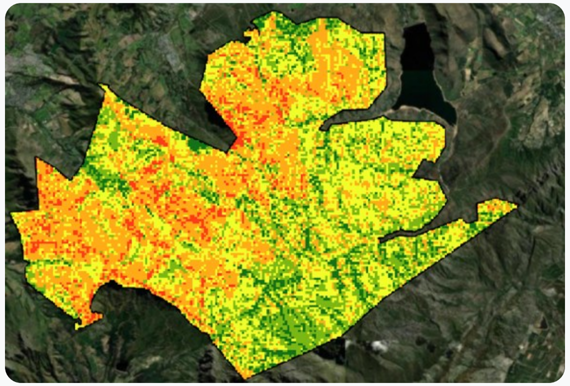
Moderate aggregation; starts to lose fine-grained environmental noise.



30m

Spatial Resolution

Standard resolution for environmental modeling and MCE comparison.

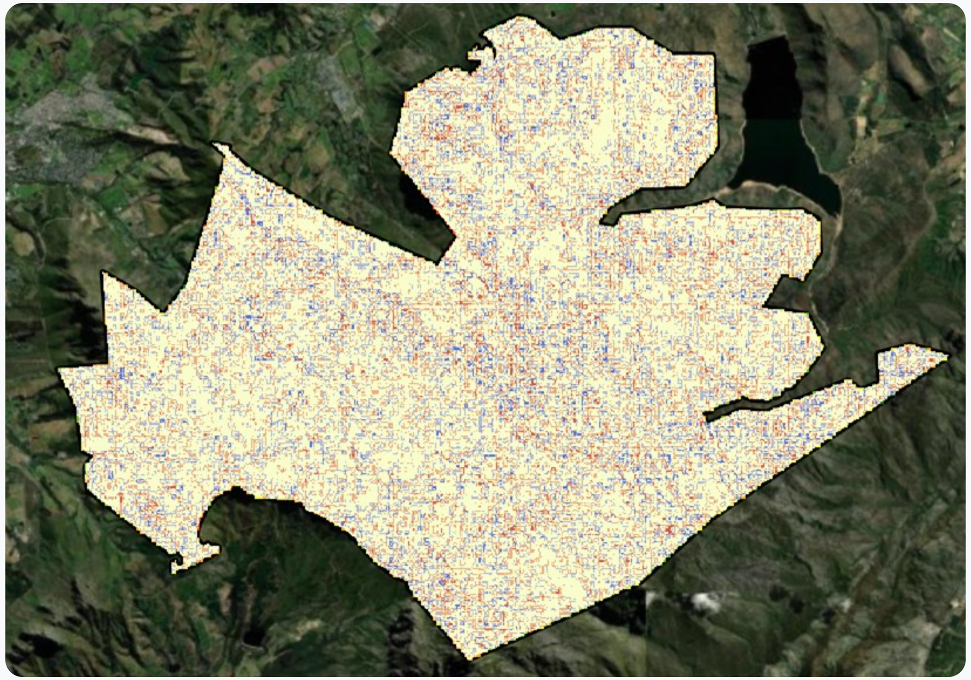


90m

Spatial Resolution

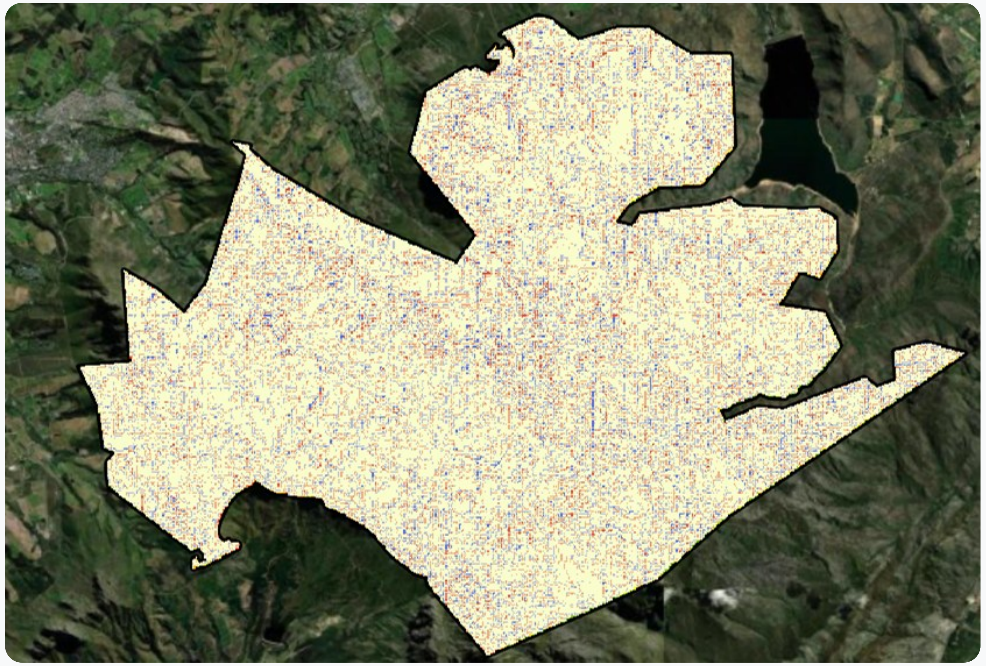
Coarse resolution; generalizes features and increases uncertainty.

Sensitivity and uncertainty analysis
- Spatial Resolution Quantified



30m vs 60m

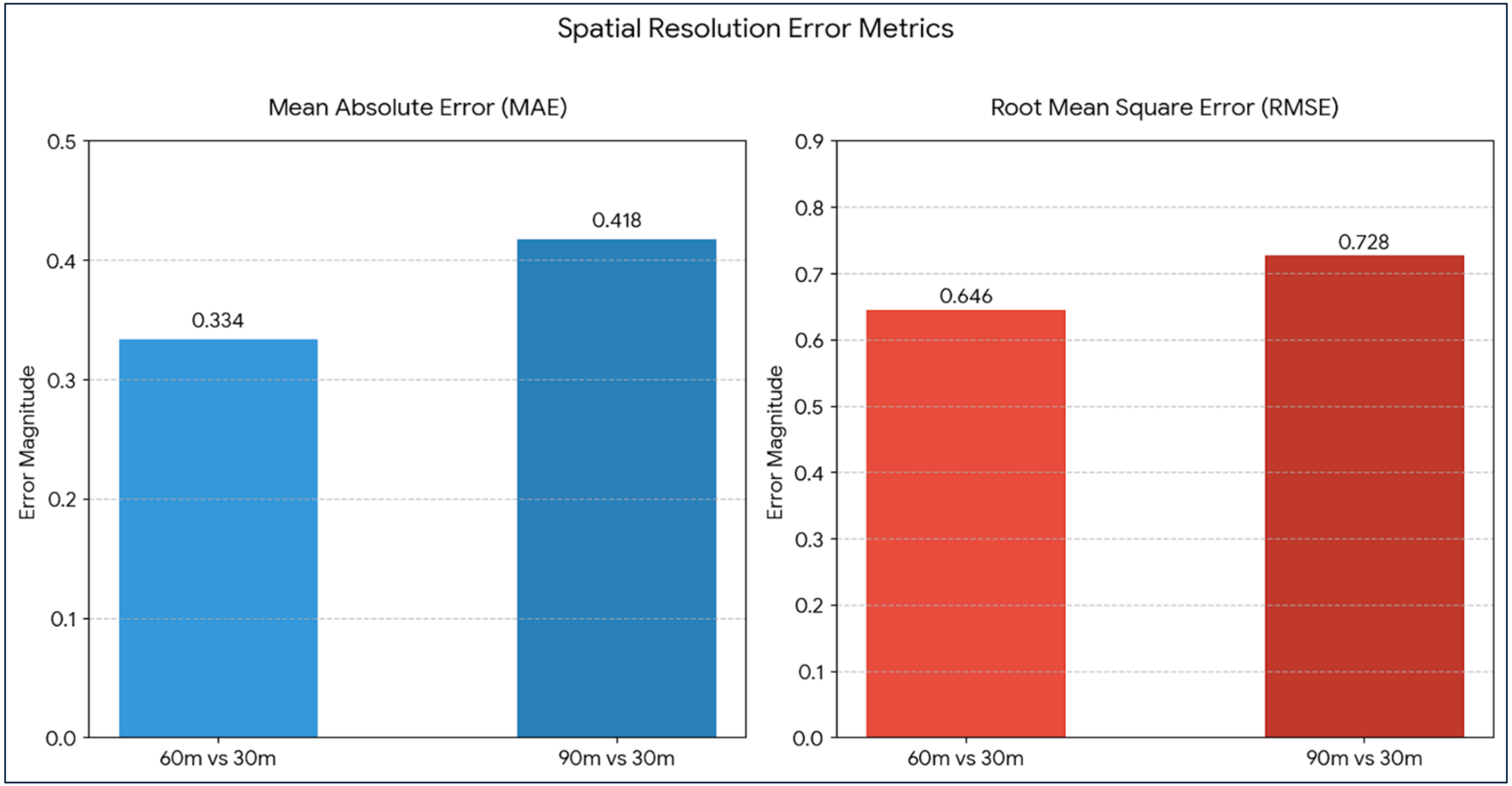
Difference Mapping



60m vs 90m

Difference Mapping

Sensitivity and uncertainty analysis - Spatial Resolution Quantified



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